MILITARY SPECIFICATION

POWER SUPPLY ASSEMBLY

LD546974

(For Afterbody Test Set MK 442 Mod 0 and
Electrical Cable Test Set MK 444 Mod 0)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy.

1. SCOPE

1.1 This specification covers the requirements for the procurement of the Power Supply Assembly, LD546974 and BuWeps Dwg 2159222, herein referred to as "assembly." (See 6.1 for description and intended use.)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-636	Boxes, Fiber

Military

MIL-P-116	Preservation, Methods of
MIL-S-901	Shockproof Equipment, Class HI (High Impact), Shipboard Application, Tests for
MIL-E-16400	Electronic Equipment, Naval Ship and Shore: General Specifications

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STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-167	Mechanical Vibrations of Shipboard Equipment

DRAWINGS

Bureau of Naval Weapons

LD564974	Power Supply Assembly
2159222	Power Supply Assembly
LD535768	Test Set, Afterbody, Torpedo MK 442 Mod 0
LD535760	Test Set, Electrical Cable, MK 444 Mod 0

(Copies of documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified in the contract or order, the preproduction sample shall:

(a) Be manufactured using the procedures proposed for the production lot.

(b) Consist of one assembly.

(c) Be tested as specified in Section 4 herein and is for the purpose of determining that, prior to starting production, the supplier's production methods are capable of yielding items that comply with the technical requirements of the contract.

3.2 Basic requirements

3.2.1 Contractor-furnished data. Data shall be supplied for preproduction and periodic production samples; however, no data is to be furnished with the deliverable production lot, unless specified herein or in the contract or order.

3.2.2 Materials and treatments of materials and parts. Unless otherwise specified herein, or in the contract or order, all materials and treatments of materials and parts used in the manufacture of the assembly shall be in accordance with LD 546974. Fungus nutrient material shall not be used.

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3.3 Performance requirements. The assembly shall meet the following performance requirements when subjected to the tests of 4.5, 4.6, and 4.7.

3.3.1 Input voltage. The assembly shall operate with a 38 ±3 root-mean-square (rms) 60 cps voltage applied between J1-B and J1-C.

3.3.2 Output voltage. The assembly shall provide the following voltages:

(a) A regulated +30.0 ±0.3 vdc output voltage, with or without a 20-ohm load, at J1-L (positive) and J1-K (common).

(b) An unregulated +50 ±5 vdc, with no load, at J1-D (positive) and J1-K (common).

3.3.3 Resistance. The resistance from the assembly test points to the pins of assembly connector P1 shall be as follows:

Assembly Test Point	To Connector P1 Pin	Resistance (Ohms)
TP1	P1-K	0.5 max
TP2	P1-L	0.5 max
TP3	P1-D	0.5 max

3.4 Environmental requirements. Unless otherwise specified the assembly, when installed in the test set, shall be capable of conforming to the performance and product characteristics of this specification during and after exposure, at any temperature in the applicable range, to the environmental requirements under all conditions specified in this section.

3.4.1 Vibration (see 6.3.1). When rigidly mounted the assembly shall be capable of withstanding the type I vibration specified in Standard MIL-STD-167, except that the variable frequency and endurance test requirements shall be as follows: The assembly in operating condition shall be vibrated from 5 to 15 cps at a half peak amplitude of 0.030 ±0.006 inch and from 16 to 25 cps at a half peak amplitude of 0.020 ±0.004 inch. The vibration shall be applied in discrete intervals of 1 cps and applied for one minute at each interval.

3.4.2 Shock (see 6.3.2). When rigidly mounted the assembly shall be capable of withstanding the shock required by Specification MIL-S-901, for Grade B, Class I, lightweight equipment, Type A, in the nonoperating condition, except that the height of the hammer drops shall be: 1 foot, 1-1/2 feet and 3 feet.

3.4.3 Temperature. The assembly shall be capable of withstanding the temperatures specified in Specification MIL-E-16400, for Class 4 equipment.

3.4.4 Humidity. The assembly shall be capable of withstanding the humidity requirements specified in Specification MIL-E-16400.

3.5 Service life. The service life of the assembly shall be 10,000 hours of operation.

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3.6 Identification. The assembly shall be permanently identified in accordance with applicable documents listed in LD 546974.

3.7 Workmanship. The workmanship and finish shall be of sufficiently high grade to assure satisfactory operation consistent with the requirements of this specification. All parts used in the assembly shall be thoroughly clean and free of excess materials, chips, and loose or splattered foreign materials. The standards of workmanship exhibited in any approved preproduction sample, shall determine the requirements of the contract relative to workmanship insofar as *not specifically covered by applicable specifications.*

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspections of the assembly shall be *classified as follows:*

(a) Preproduction inspection

(1) Preproduction sample (see 4.3.1)

(2) Preproduction tests (see 4.5)

(b) Periodic production inspection

(1) Periodic production sampling (see 4.3.2)

(2) Periodic production tests (see 4.6)

(c) Quality conformance inspection

(1) Quality conformance sampling (see 4.3.3)

(2) Quality conformance tests and examinations (see 4.7)

4.3 Sampling

4.3.1 Preproduction sample. A preproduction sample manufactured in accordance with 3.1, shall be subjected to the preproduction tests detailed in 4.5 at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the assembly by the supplier, prior to approval of the preproduction sample, shall be at the supplier's risk.

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4.3.2 Periodic production sampling. Periodic production samples shall be selected from units having satisfactorily passed quality conformance tests and examinations of 4.7. These samples shall be subjected to the tests of 4.6 at an activity designated by the procuring activity. Periodic production sampling shall be as follows:

(a) Step 1. Upon initiation of production randomly select 1 unit from each group of 20 units. Upon satisfactory compliance of samples selected from 3 consecutive groups, proceed to Step 2. Each sample so selected shall be numbered consecutively starting with Sample Number 1.

(b) Step 2. Randomly select 1 unit from each group of 50 units. Continue with Step 2 for the balance of the contract or until a defective unit is found. Each sample so selected shall be numbered consecutively starting with a sample number one greater than the highest sample number selected in Step 1.

4.3.2.1 Defective units. When a defective unit, a unit that fails to meet all the tests and examinations specified, is found in any group, acceptance of all units on hand shall be withheld and the contractor shall:

(a) Determine the cause of the defect.

(b) Submit to the procuring activity for approval, a complete failure analysis and the proposed corrective action to prevent recurrence of the defect.

(c) Submit to the procuring activity for approval the proposed retest program to verify that the corrective action eliminated the defect.

(d) Take the necessary actions to correct or eliminate the defect from all units on hand and, if so directed by the procuring activity, from those units previously shipped.

(e) Return to Step 1 (4.3.2.(a)) and proceed as indicated when inspection is resumed.

4.3.3 Quality conformance sampling. The tests and examinations specified in 4.7 shall be conducted on each assembly to be delivered, therefore, there shall be no sampling plan for acceptance. Defective units shall be rejected.

4.4 Test conditions. Unless otherwise specified, the assembly shall be subjected to tests under the following conditions:

(a)	Temperature	Room ambient (60 to 95 degrees F)
(b)	Humidity (Relative)	Room ambient, up to 95 percent
(c)	Altitude	Normal ground atmospheric pressure

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4.5 Preproduction tests. The preproduction sample shall be subjected to the environmental tests of 4.5.2. Prior to and immediately after the preproduction tests, the assembly shall be subjected to, and satisfactorily complete, the quality conformance tests and examinations of 4.7. Certified inspection data covering the results of preproduction tests shall be forwarded to the procuring activity. Certified inspection data shall consist of the following:

(a) Date of tests and names of persons conducting tests.

(b) Test plan.

(c) Serial number of the assembly.

(d) Results of preproduction tests (4.5.2).

4.5.1 Preproduction test equipment. The following items of test equipment are required to perform the preproduction tests specified herein:

(a) A vibration table capable of applying vibration as specified herein.

(b) A shock testing machine capable of applying shocks specified herein.

(c) A humidity-temperature chamber capable of applying the humidity, temperature, and life tests specified herein.

(d) Mounting fixture as shown in figure 1.

(e) Items of test equipment specified in table I.

4.5.2 Environmental tests

4.5.2.1 Vibration. Install the assembly in the mounting fixture and rigidly attach the mounting fixture to the vibration table. Vibrate the assembly in accordance with vibration schedules contained in Standard MIL-STD-167, with exceptions specified in 3.4.1.

4.5.2.2 Shock. Install the assembly in the mounting fixture and rigidly attach the mounting fixture to the shock test machine. Subject the assembly to the mechanical shock schedule contained in Specification MIL-S-901, with exceptions as specified in 3.4.2.

4.5.2.3 Temperature. Subject the assembly to temperature tests, both operating and nonoperating, as specified in Specification MIL-E-16400 for Class 4 equipment.

4.5.2.4 Humidity. Subject the assembly to the humidity tests contained in Specification MIL-E-16400.

4.5.2.5 Accelerated life. Subject the assembly to the accelerated life test of MIL-E-16400, Class 4 equipment.

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NOTE:
USE 5/16 INCH ALUMINUM

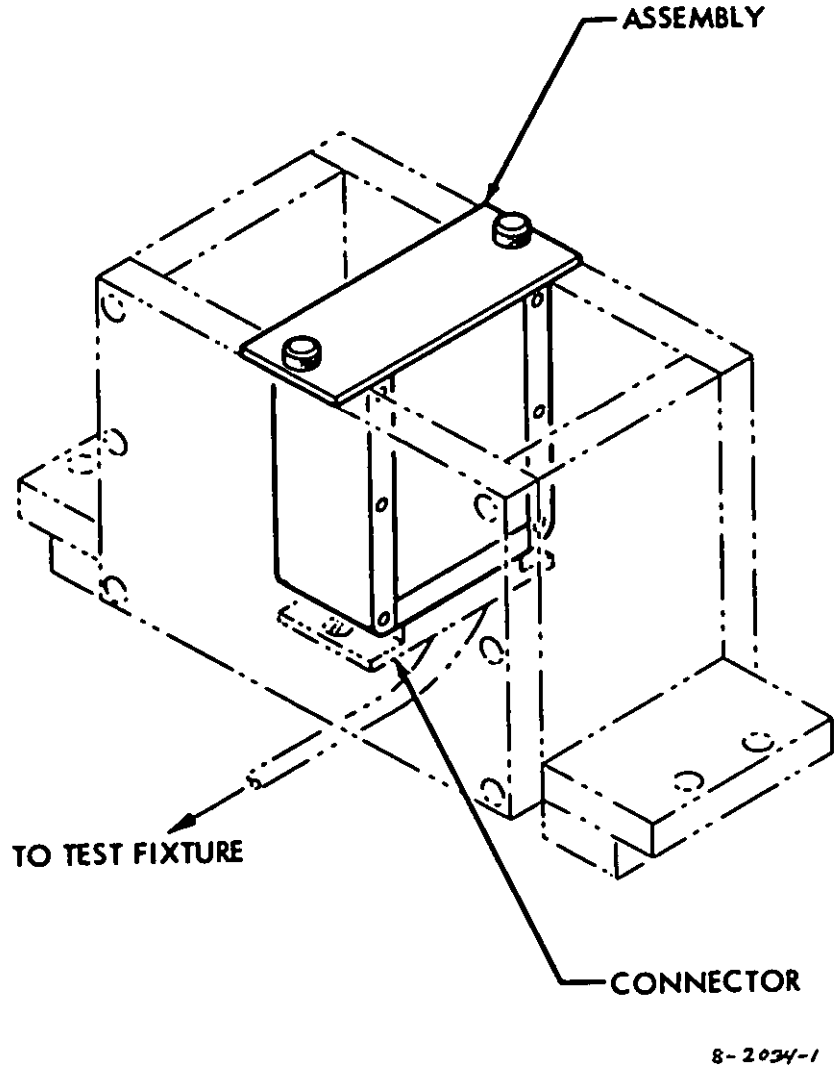


Figure 1. Power Supply Assembly, Environmental Test Mounting Fixture

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4.6 Periodic production tests

4.6.1 Periodic production test equipment. The items of test equipment, specified in 4.5.1, will be required to perform the tests specified herein:

NOTE

Periodic production samples that are to be subjected to accelerated life tests of 4.5.2.5 need not be subjected to the temperature tests of 4.5.2.3.

4.6.2 Periodic production tests. After satisfactorily passing the quality conformance tests and examinations (4.7), each odd-numbered periodic production sample shall be subjected to vibration tests (4.5.2.1) and temperature tests (4.5.2.3), and each even-numbered periodic production sample shall be subjected to shock tests (4.5.2.2) and humidity tests (4.5.2.4). In addition to the vibration and temperature tests, the first, the fifth, and every fifth periodic production sample thereafter shall be subjected to accelerated life tests (4.5.2.5). Unless otherwise specified, the sample shall be deemed a defective unit if, subsequent to the periodic production tests, it does not pass the quality conformance tests and examinations.

4.7 Quality conformance tests and examinations. Unless otherwise specified, each assembly delivered under contract shall be subjected to the tests and examinations specified herein. A report form is provided in the appendix for recording quality conformance test data.

4.7.1 Acceptance criteria. Failure of the end product to pass any examination or test of this section will automatically classify the unit as nonconforming. When nonconforming units are corrected by the contractor, the unit shall be reinspected. The reinspection of a nonconforming unit may be limited to the test or examination which defined the nonconformance or, when so directed by the cognizant inspector, a complete re-examination and retest of the unit may be required. Nonconforming units, which have not been corrected, will be considered for acceptance only upon formal application by the contractor to the procuring activity via the cognizant Government representative.

4.7.2 Quality conformance test equipment. Items of test equipment listed in table I are required to perform quality conformance tests specified herein. All items of test equipment shall be of sufficient accuracy and precision to permit valid performance of required tests. Test instruments shall be calibrated at regular intervals. Procedures for and results of all calibrations shall be subject to approval of the cognizant Government Inspector. Items having equivalent characteristics may be substituted where necessary. Verify that all test equipment used is identified as being within calibration.

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Table I

Quality Conformance Test Equipment

Item	Identification
Digital Voltmeter (Range: 0.001 to 999.9 volts dc or rms; DC Accuracy: ±0.05% of reading ±1 digit; AC Accuracy: ±0.1% of reading ±2 counts.)	Hewlett-Packard Model 3440A with 3445A AC/DC Range Unit
Transformer, Variable (Range: 0 to 120 vac at 2 amperes.)	Variac Model W2
Multimeter (Resistance range: 0 to 20 megohms; Accuracy: within 3 degrees of arc on resistance scale.)	Simpson Model 260

Power Supply Assembly Test Fixture
(consisting of the following components) (See figure 2)

Circuit Symbol	Part	Designation	Quantity
J1	Connector	MS3116F14-12S	1
R1	Resistor, 20 ohm	RE75G20R0	1
S1	Switch SPDT	MS35058-22	1
TP2	Binding Post Blk	Superior DF30BC	1
TP1, TP3, TP4, TP5	Binding Post Red	Superior DF30RC	4

4.7.3 Test procedure.

4.7.3.1 Test setup.

4.7.3.1.1 Turn on and adjust variable transformer for a zero indication on digital voltmeter.

4.7.3.1.2 Connect digital voltmeter between TP4 and TP5 of test fixture shown in figure 2. Connect assembly to the power supply assembly test fixture, and to the variable transformer as shown in figure 2.

4.7.3.1.3 Adjust variable transformer for a 38 ±3 vrms indication on digital voltmeter.

4.7.3.2 Regulated output voltage test.

4.7.3.2.1 Connect digital voltmeter between TP1 (+30 V REG) and TP2 (common). Set switch S1 to OFF position. Digital voltmeter shall indicate +30 ±0.3 vdc.

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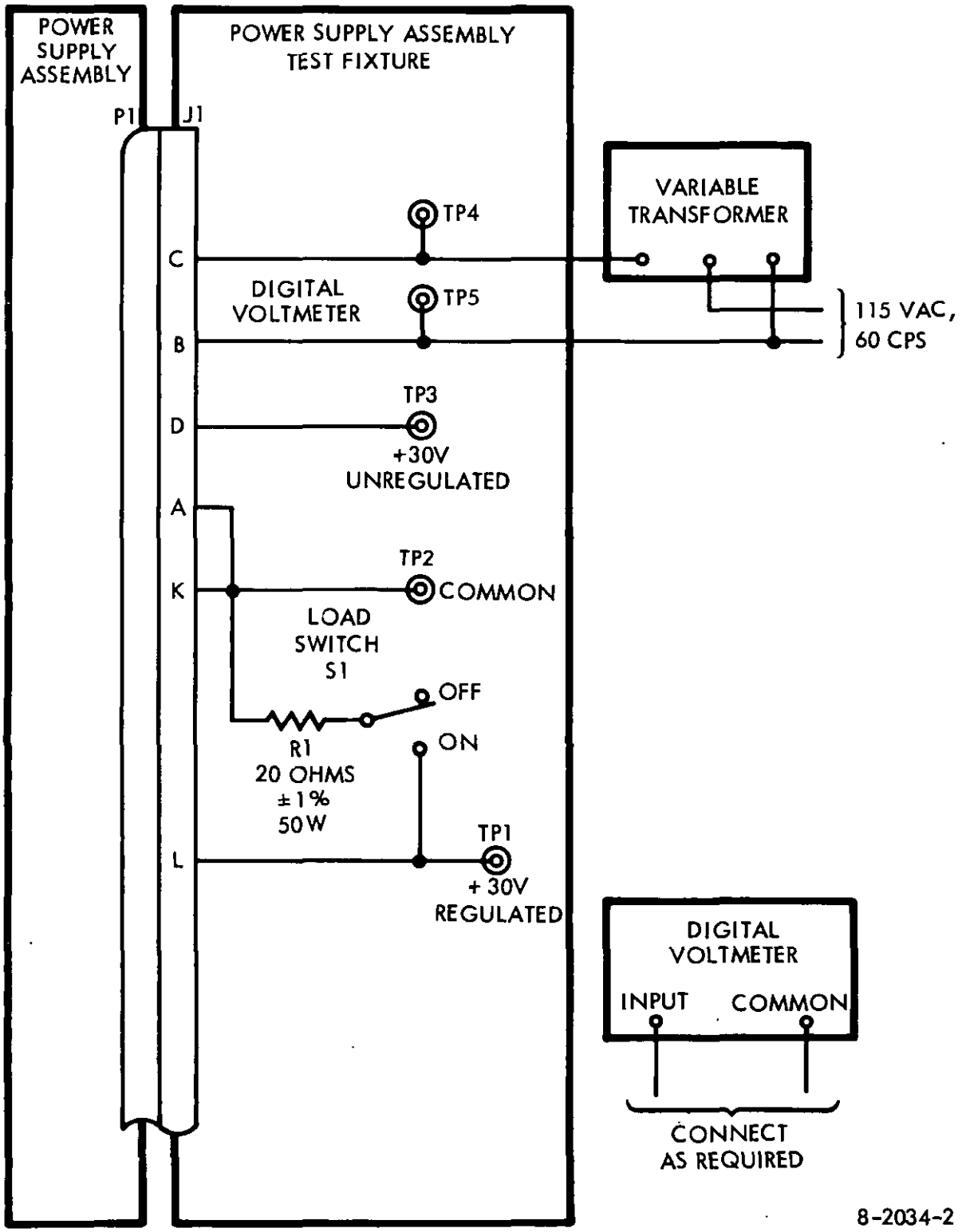


Figure 2. Power Supply Assembly, Test Setup

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4.7.3.2.2 Set switch S1 to ON position. Digital voltmeter shall indicate +30.0 ±0.3 vdc.

4.7.3.3 Unregulated output voltage test.

4.7.3.3.1 Set switch S1 to OFF position, and connect digital voltmeter between TP3 (+30 V UNREG) and TP2 (common). Digital voltmeter shall indicate +50.0 ±0.4 vdc.

4.7.3.4 Resistance measurements.

4.7.3.4.1 Turn off variable transformer and remove assembly from test fixture, then measure resistance between assembly test points and pins of connector P1 as follows:

Assembly Test Point	To Connector P1 Pin	Resistance (Ohms)
TP1	K	0.5 max
TP2	L	0.5 max
TP3	D	0.5 max

4.7.4 Examination of product. The assembly shall be examined for conformance to the requirements specified herein and on the drawings listed in LD546974. The presence of any one of the following defects shall constitute a failure.

(a) Assembly not complete; components or parts missing.

(b) Unauthorized material used or unauthorized repairs made.

(c) Dents, scrapes, nicks, bent pins, poor electrical connections, or any other damage as determined by the procuring activity.

(d) Receptacles do not accept mating plugs.

(e) Identification markings missing, located incorrectly, illegible or incomplete.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be Level A or C as specified (see 6.2).

5.1.1 Level A

5.1.1.1 Cleaning. Cleaning, if necessary, shall be in accordance with process C1; and drying, if required, shall be in accordance with procedure D-1 or D-4 of MIL-P-116.

5.1.1.2 Unit packaging. Unit packaging of the assemblies shall be in accordance with Method IIb of Specification MIL-P-116. The unit package shall contain one assembly in fiberboard containers. The outer container shall be in accordance with Specification PPP-B-636; Style RSC; Type CF; Class - weather resistant; Variety SW; Grade V3c.

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5.1.1.3 Intermediate package. Not applicable.

5.1.1.4 Cushioning. The unit package, and pack, shall incorporate sufficient cushioning material, bracing, or other adequate shock absorbing devices, to assure that the assembly will meet all the requirements of Section 3 after transportation. Cushioning material and packaging procedure shall be such that the assembly shall not be contaminated with dust or foreign materials. At the time of packaging or packing, the moisture content of the cushioning material shall be no more than 5 percent. The pH range of the material shall be between 6.0 and 8.0.

5.1.2 Level B. Not applicable.

5.1.3 Level C. Cleaning, drying, preservation and packing shall be in accordance with the supplier's commercial practice when such practices are approved by the procuring activity.

5.2 Packing

5.2.1 Level A

5.2.1.1 Exterior containers. Assemblies, packaged as described in 5.1.1.2, shall be packed in a fiberboard container in accordance with Specification PPP-B-636; Style RSC; Type CF; Class - Weather resistant; Variety SW; Grade V3c. Exterior containers shall contain the number of unit packages specified in the contract or order (see 6.2) or, in the event no quantity is specified, the number of unit packages packed shall be at the option of the contractor.

5.2.2 Level B. Not applicable.

5.2.3 Level C. Assemblies, packaged in accordance with 5.1, shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Shipping containers shall conform to the carrier rules and regulations applicable to the mode of transportation.

5.3 Marking

5.3.1 Normal markings. In addition to the markings required by contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129, including serial numbers.

6. NOTES

6.1 Description and intended use

6.1.1 Description. The assembly provides a regulated +30 vdc and an unregulated +30 vdc.

6.1.2 Intended use. The assembly is an integral part of the Afterbody Test Set Mark 442 Mod 0, used to verify compliance with performance requirements of the Torpedo Mark 46 at Fleet level.

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6.2 Ordering data. Procurement documents shall specify the following:

(a) Title, number and date of this specification.

(b) Required level of preservation and packaging, and packing.

(c) Packing quantity.

(d) Assigned activity for preproduction and periodic tests.

(e) Any exceptions to this specification.

(f) Provisions for additional samples required because of failure of any sample(s) to pass the prescribed inspections.

(g) Required data.

6.3 Definitions

6.3.1 Vibration. Vibration is a sinusoidal motion defined by amplitude and frequency. The amplitude may refer to displacement or acceleration. The displacement is given in one half the peak-to-peak excursion in inches or the corresponding maximum acceleration in gravity units (g). The frequency is given in cycles per second (cps).

6.3.2 Shock. Shock is an abrupt change in the position of a body resulting from the sudden application of a large external force of short duration. Shock is described as a pulse force in units of gravitational acceleration (g's), of duration in units of time, milliseconds (ms), and a shock wave approximating a one-half sine wave. The time duration shall be measured between two points where two lines intersect the zero "g" line. These lines are tangent to the shock wave slopes.

6.4 Preparation for delivery criteria. Criteria for use of the proper level of preservation, packaging, and packing shall be as follows:

Level A	This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.
Level B	Not applicable.
Level C	This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

Custodian:

Navy - NOSC

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REPORT FORM

QUALITY CONFORMANCE TESTS AND EXAMINATIONS

FOR

POWER SUPPLY ASSEMBLY

LD 546974

(For Afterbody Test Set MK 442 Mod 0)

EQUIPMENT STATUS

CONTRACTOR CERTIFICATION		GOVERNMENT ACCEPTANCE	
Contractor		Date of Acceptance	
Contract		Accepted by	
Equipment Ser. No.		Title	
Insp. Req. Satisfied		Agency	

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REPORT FORM

QUALITY CONFORMANCE TEST AND EXAMINATION

10.0 Quality conformance test and examination

10.1 Scope. This report form contains data sheets to record the results of the quality conformance tests and examinations of 4.7 of this specification.

10.2 Verification. Where a limit or limits are provided, the value of a specific reading obtained shall be recorded. If limits are not provided, satisfactory compliance of the requirement must be verified.

10.3 Test results.

		Reference	Limits	Reading
10.3.1	Test equipment	4.7.2	In calibration	_____
10.3.2	Test procedure	4.7.3		
10.3.2.1	Regulated output voltage	4.7.3.2		
	Voltage (no load)	4.7.3.2.1	+29.7 to +30.3 vdc	_____
	Voltage (with load)	4.7.3.2.2	+29.7 to +30.3 vdc	_____
10.3.2.2	Unregulated output voltage	4.7.3.3		
	Voltage	4.7.3.3.1	+49.6 to +50.4 vdc	_____
10.3.2.3	Resistance measurements	4.7.3.4		
	Test Point To Pin P1	4.7.3.4.1		
	TP1 K		0.5 ohm max	_____
	TP2 L		0.5 ohm max	_____
	TP3 D		0.5 ohm max	_____
10.4	Examination of product	4.7.4	Satisfactory	_____
10.5	Preservation and packaging	5.1	Satisfactory	_____

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-P-82563(OS) POWER SUPPLY ASSEMBLY LD546974 (For Afterbody Test Set Mk 442 Mod 0 and ~~Electrical Cable Test Set Mk 444 Mod 0~~)

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426

C-8979

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